

$$1. \begin{cases} A_1 x = b_1 \\ A_2 x = b_2 \end{cases}$$

$$x = [1, 0]^T$$

(a) Change in Solution when b is perturbed

Problem states that we add 0.01 to the first component of the rhs

$$\rightarrow \Delta b = [0.01, 0]^T$$

Thus, we are solving perturbed system

$$A(x + \Delta x) = b + \Delta b, \text{ where } \Delta b = [0.01, 0]^T$$

Since A is invertible:

$$\Delta x = A^{-1} \Delta b$$

So we need to compute A^{-1} , and then multiply Δb

Computed results:

$$A_1^{-1} = \begin{bmatrix} 0.073 & -0.0686 \\ 0.0742 & 0.092 \end{bmatrix}$$

$$A_2^{-1} = \begin{bmatrix} 30.0 & 36.0 \\ 22.0 & 28.0 \end{bmatrix}$$

• Case A_1

$$\Delta x_1 \approx 10^{-3} \begin{bmatrix} 0.7305 \\ 0.7413 \end{bmatrix}$$

$\rightarrow$ changes by about 10^{-3}

• Case A_2

$$\Delta x_2 \approx \begin{bmatrix} 0.30 \\ -0.22 \end{bmatrix}$$

$\rightarrow$ change is about 0.3

So A_2 produces a much larger change in x for the same small change in b .

(b) Condition Numbers

$$\kappa(A) = \|A\| \cdot \|A^{-1}\|$$

$\rightarrow$ I'll use ∞ norm since it's easier to compute with.

$$\kappa_{\infty}(A_1) = \|A_1\|_{\infty} \cdot \|A_1^{-1}\|_{\infty} \approx 2.12$$

$$\kappa_{\infty}(A_2) = \|A_2\|_{\infty} \cdot \|A_2^{-1}\|_{\infty} \approx 1123.0$$

(c) Relationship

$\kappa(A)$ show how well/ill conditioned the matrix is.

if $\kappa(A)$ is large:

- matrix is ill-conditional
- Small changes of A or b makes $\uparrow \Delta x$

From above questions, A_1 has small $\kappa \rightarrow \downarrow \Delta x$ (order 10^{-3})

A_2 has large $\kappa \rightarrow \uparrow \Delta x$ (order 10^{-1})

$\therefore$ The larger Condition number, the more sensitive the solution is to perturbations in b .

(d) Upper bound on rel. error

$$A(x+\Delta x) = b+\Delta b$$

$$Ax + A\Delta x = b + \Delta b, \text{ but } Ax = b$$

$$A\Delta x = \Delta b$$

$$\Delta x = A^{-1} \Delta b$$

$$\|\Delta x\| \leq \|A^{-1}\| \cdot \|\Delta b\|, \text{ again } Ax=b \Leftrightarrow b=Ax$$

$$\text{Thus } \|b\| = \|Ax\| \leq \|A\| \cdot \|x\| \Leftrightarrow \frac{\|b\|}{\|x\|} \leq \|A\| \Leftrightarrow \frac{1}{\|x\|} \leq \frac{\|A\|}{\|b\|}$$

Substitute:

$$\frac{\|\Delta x\|}{\|x\|} \leq \|A^{-1}\| \cdot \|\Delta b\| \frac{\|A\|}{\|b\|}$$

$$K(A) = \|A\| \cdot \|A^{-1}\|$$

$$\rightarrow \frac{\|\Delta x\|}{\|x\|} \leq K(A) \cdot \frac{\|\Delta b\|}{\|b\|}$$

2.
$$A = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 3 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 3 & -1 & 0 & -1 & 0 \\ 0 & -1 & -1 & 3 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 2 & 0 & 0 \\ -1 & 0 & -1 & 0 & 0 & 4 & -1 \\ -1 & 0 & 0 & 0 & 0 & -1 & 3 \end{bmatrix}$$

(a) Find $A=LL^T$

→ Cholesky Fact.

$$\text{for } j=1 \dots n \quad 1) L_{jj} = \sqrt{A_{jj} - \sum_{k=1}^{j-1} L_{jk}^2}$$

2) for $i > j$:

$$L_{ij} = \frac{1}{L_{jj}} \left(A_{ij} - \sum_{k=1}^{j-1} L_{ik} L_{jk} \right)$$

diagonal entries start with $\sqrt{3}$ ($A_{11}=A_{22}=A_{33}=3$)

Compute with the algorithm, we get

$$L = \begin{bmatrix} \sqrt{3} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sqrt{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \sqrt{\frac{7}{3}} & 0 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{3}} & 0 & -\frac{\sqrt{3}}{7} & \sqrt{\frac{162}{100}} & 0 & 0 \\ -\frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{3}} & -\frac{\sqrt{3}}{7} & -\frac{37}{98} & \sqrt{\frac{3284}{1000}} & 0 \\ -\frac{1}{\sqrt{3}} & 0 & 0 & 0 & -\frac{7357}{10000} & \sqrt{\frac{27257}{10000}} & 0 \end{bmatrix} \approx \begin{bmatrix} 1.7321 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1.7321 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1.7321 & 0 & 0 & 0 & 0 \\ 0 & -0.5774 & -0.5774 & 1.5275 & 0 & 0 & 0 \\ 0 & -0.5774 & 0 & -0.2182 & 1.2524 & 0 & 0 \\ -0.5774 & 0 & -0.5774 & -0.2182 & -0.0374 & 1.8123 & 0 \\ -0.5774 & 0 & 0 & 0 & 0 & -0.7357 & 1.4579 \end{bmatrix}$$

(b) Finding permutation P for sparser L_1

Step 1: Represent A as a graph

- Each row/col i is a node
- If $A_{ij} \neq 0$ for $i \neq j$, there is an edge $i-j$
- undirected graph

Step 2: Apply Minimum Degree (Greedy algorithm that finds Smallest degree node)

[Algorithm]

1. Compute the degree of each node in the current graph
2. Select node with Smallest degree
3. Remove that node from graph, but connect all of its neighbors
4. Record removal order

→ reverse of the removal order gives permutation p.

Step 3: Permutation matrix P

from step 1~2, i got Removal order: [5, 2, 4, 3, 1, 6, 7] (but this order can be differ)

∴ reverse: [7, 6, 1, 3, 4, 2, 5] = p

Step 4: Build P^TAP

$$P = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad P^TAP = \begin{bmatrix} 3 & -1 & 0 & 0 & 0 & -1 \\ -1 & 4 & -1 & 0 & 0 & 0 \\ 0 & -1 & 3 & 0 & 0 & -1 \\ 0 & 0 & 0 & 3 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 3 & 0 \\ -1 & 0 & -1 & 0 & 0 & 3 \end{bmatrix}$$

⇒ Compute P^TAP

$$P^TAP = L_1 L_1^T \quad L_1 = \begin{bmatrix} 1.7321 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.5774 & 1.9149 & 0 & 0 & 0 & 0 & 0 \\ 0 & -0.5222 & 1.6432 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1.7321 & 0 & 0 & 0 \\ 0 & 0 & 0 & -0.5774 & 1.2916 & 0 & 0 \\ 0 & 0 & -0.6086 & -0.5774 & -0.4472 & 1.9416 & 0 \\ -0.5774 & -0.1741 & -0.6086 & 0 & 0 & 0.226 & 1.5119 \end{bmatrix}$$